

2.5 Integration

Pure Mathematics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Integration by substitution identifies an inner expression whose derivative also occurs. Area of a region between graphs is integral of upper minus lower curve; split at intersections. For volumes of revolution about an axis use $\pi \int y^2 dx$ when permitted by syllabus.

Worked example: follow the reasoning

$\int 2x(x^2+1)^3 dx = (x^2+1)^4/4 + C$ via $u = x^2+1$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Evaluate $\int$ from 0 to 1 of $2x(x^2+1) dx$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Substitute $u=x^2+1$, $du=2x \, dx$ [1]; integrate u from 1 to 2 [1]; $[u^2/2]_1^2=3/2$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.